Neural Network-Inspired Phase-Coded Waveform Design for MIMO Radar Based on Gradient Descent

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Peak sidelobe level (PSL) stands as a critical performance metric in the design of multiple-input multiple-output (MIMO) radar waveforms. However, optimizing PSL is a challenging task due to its high dimensionality and nonconvex nature. Traditional optimization algorithms often require sophisticated techniques with high computation complexity while the resulting PSL is relatively high. Drawing inspiration from neural network optimization, this article presents an efficient approach that employs gradient descent (GD) to design low sidelobe phase-coded waveforms for MIMO radar. This is accomplished by smoothing the PSL's maximum function with a Log-Sum-Exp (LSE) function, which serves as the objective function for the waveform design problem. Specifically, the new LSE function controls the degree of approximation to the maximum function, preventing numerical overflow and maintaining computational accuracy. The ensuing unconstrained approximate minimization problem is amenable to GD optimization. Besides the new LSE objective function, another key contribution lies in combining GD with neural network optimization, resulting in a significantly faster optimization process compared to traditional methods. Utilizing neural network frameworks, the GD algorithm benefits from automatic differentiation and GPU acceleration, enabling efficient optimization of large waveform sets. Extensive numerical studies demonstrate that the proposed method can design waveform sets with low PSL or weighted PSL effectively, which can be closer to the Welch bound as compared to conventional approaches.

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I. INTRODUCTION

Compared to traditional single-input single-output (SISO) radar and phased array radar, multiple-input multiple-output (MIMO) radar presents a higher degree of freedom in transmit waveforms [1]. Waveform diversity technology can significantly improve the performance of MIMO radar in such applications as target detection [2], tracking [3], imaging [4], [5], jamming suppression, and clutter mitigation [6], [7], [8]. The design of orthogonal waveform sets is a prerequisite for achieving waveform diversity gain in MIMO radar. Considering practical applications, constant modulus is the primary constraint in waveform design [9].

Orthogonal waveform sets can be achieved through three approaches: time-division, frequency-division, and code-division. Both time-division and frequency-division offer nearly perfect orthogonality. However, time-division results in a loss of transmit power, while frequency-division leads to high-range side lobes [10]. In contrast, code-division provides a large time-bandwidth product and greater resistance to interference compared to the previous two methods [11], [12]. Due to the limited time-bandwidth of actual radar signals, it is not possible to design completely orthogonal phase-coded waveform sets [13]. Therefore, practical implementations can only utilize approximately orthogonal waveform sets.

Aperiodic auto- and cross-correlation are indicative of the orthogonality of phase-coded waveform sets [14], [15]. Henceforth, when referring to correlation, we imply aperiodic correlation without explicitly mentioning "aperiodic." The standard metrics used to assess the quality of auto- and cross-correlation within these waveform sets are the peak sidelobe level (PSL) [16] and integrated sidelobe level (ISL) [17]. In addition, there are some more refined metrics, such as peak auto-correlation sidelobe ratio (PASR) and peak cross-correlation ratio (PCCR). PCRR is also called peak correlation level (PCL). Although the abbreviations are different, they essentially are equivalent to PSL. For the suppression of certain region of the auto/cross-correlation functions, modified merit factor [18] and modified PCL [19] also were mentioned. In this article, normalized PSL and ISL are utilized as metrics, each of which has theoretical lower bounds [20], [21].

Several algorithms have been developed to optimize ISL, approaching its lower bound. The cyclic algorithm new

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(CAN) [18] and the extended multi-CAN [22] use cyclic procedure to design waveforms approaching ISL lower bound in frequency domain, accelerated with fast Fourier transform (FFT) based operation. Subsequently, the authors in [23], [24], [25], and [26] used the majorization minimization (MM) method to optimize ISL, with the main goal of accelerating the convergence speed of large-scale waveforms. Compared to MM-Corr [25], ISL-New algorithm [26] utilizes different majorization functions, resulting in a lower or comparable computational burden per iteration and faster convergence speed. These algorithms can achieve ISL values close to the lower bound, but the PSL values remain relatively high.

To optimize PSL, an optimization problem is typically formulated with PSL as the objective function. This approach is analogous to the optimization of ISL. However, the optimization of PSL differs slightly from ISL, as it is a nonsmooth minimax problem. The p-norm model was used to tackle this minimax problem [27], [28], [29], [30]. When $p \to 2$, the expression is close to ISL, and when $p \to \infty$, it approximates PSL. In [27], a compact expression for the gradients of the p-norm model of SISO waveforms was proposed, which reduces the computational complexity of the gradients from $\mathcal{O}(N^3)$ to $\mathcal{O}(N \log N)$. The dual problem of the PSL optimization problem was solved using the Lagrange multiplier method, resulting in lower PSL but ISL relatively high [28]. In [30], an optimization framework based on block successive upper bound minimization was proposed to solve the p-norm model, which obtains near-optimal PSL and ISL values so far.

In addition to the p-norm model, there are also some literatures of optimizing continuous or binary sequences with low PSL, using the alternating projection [31], the coordinate descent method [16], [32], [33], [34], cyclic algorithm [35], and so on. These are all traditional optimization algorithms.

Various heuristic optimization algorithms have been employed for MIMO radar waveform set design, including genetic algorithms [36] and greedy algorithms [37]. However, these algorithms tend to have longer runtimes when applied to long sequences [36], [37], [38], [39]. With the rapid advancements in neural networks within the field of artificial intelligence, some researchers have explored solving waveform set design problems using neural networks [40], [41], [42], [43], [44], [45], [46]. A comprehensive optimization network (CON) [46] based on residual blocks was trained through unsupervised learning to minimize a loss function that combines the weighted sum of PSL and ISL. This approach yielded favorable results, particularly when dealing with long sequences. It is worth noting that while the results from [46] suggest the feasibility of using neural networks for waveform set design, there is no compelling theoretical basis for this approach. In the context of waveform set design problems, terms like "samples" and "training," commonly used in classification recognition, do not have direct counterparts.

Neural networks and gradient descent (GD) share similarities in waveform design [27], [30], [47], [48], [49], [50], as both methods employ GD to seek local optimal solutions. Inspired by the application of GD in solving large-scale nonconvex optimization problems in neural networks, this article introduces a GD-based algorithm for designing orthogonal waveform sets with lower PSL values. The contribution of this article is threefold as follows.

1) Different from the p-norm model, we employ the Log-Sum-Exp (LSE) function as a smooth approximation of the max function [51], [52], [53], thereby creating a smooth objective function to replace the nondifferentiable PSL. Meanwhile, the LSE trick can be utilized to avoid numerical overflow during the computation without losing computational accuracy.
2) Subsequently, we develop the optimization based on GD (Algorithm 1) and utilize the Adam optimizer [54] to minimize this unconstrained approximate optimization problem, reducing the likelihood of converging to a local minimum.
3) Furthermore, the connection between GD-based algorithms and neural networks optimization was clarified, enabling the leverage of backpropagation structure to implement GD-based algorithms (Algorithm 2). Mature software frameworks [55], [56], [57] can be exploited to reduce programming effort. At the same time, the automatic differentiation module within the neural network facilitates rapid computation of gradients [58], [59], [60], [61], thereby reducing computational time while maintaining equivalent PSL performance.

Numerical results demonstrate that the algorithm proposed in this article can design waveform sets with PSL values 1–2 dB lower to state-of-the-art waveform design algorithms, while achieving ISL values close to the lower bound with high computational efficiency.

The rest of this article is organized as follows. Section II introduces the signal model, performance metrics, and the model of the approximated optimization problem. Section III provides a detailed description of the GD-based algorithm. Section IV presents the connection between the proposed algorithm and neural networks. Section V presents numerical results. Finally, Section VI concludes this article.

Notation: Throughout this article, the matrices, vectors, and scalars are denoted in the form of bold uppercase, bold italic lowercase, and italic letters, respectively. The notations $\Re(\cdot)$ and $\Im(\cdot)$ denote the real and imaginary parts of a complex number. The superscripts $(\cdot)^*$, $(\cdot)^{\mathrm{T}}$, and $(\cdot)^{\mathrm{H}}$ denote conjugate, transpose, and conjugate transpose, respectively. The superscripts $(\cdot)^{\mathrm{f}}$ represents the flip order of elements. The notation $\tilde{(\cdot)}$ represents "generated" versions of variables. The superscripts in brackets $(\cdot)^{(d)}$ represents the iteration index. The symbol $\mathbf{I}_N$ denotes the $N \times N$ identity matrix. The symbols $\mathbf{0}$ and $\mathbf{1}$ denote a matrix or vector with all elements equal to 0 and 1, respectively. The symbol j denotes the complex number $\sqrt{-1}$. The Hadamard product and convolution are written as $\circ$ and $\otimes$, respectively. The function $\mathrm{vec}\{\cdot\}$ denotes the columnwise vectorization of a

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matrix; $|\cdot|$ calculates the modulus of a complex number; and $||\cdot||$ calculates the Frobenius norm of a matrix.

II. PROBLEM FORMULATION

Consider an MIMO radar with M channels, and each channel transmits a phase-coded waveform, which can be expressed as a sequence of length N. The ith sequence is

$$\mathbf{x}_i = [x_i(0), x_i(1), \ldots, x_i(k), \ldots, x_i(N-1)]^{\mathrm{T}} \in \mathbb{C}^N \quad (1)$$

where $x_i(k)$ denotes the $(k+1)$th element in the ith waveform. $\mathbb{C}^N$ denotes the N-dimensional complex number space.

Since the amplitude of the sequence under constant modulus constraint does not affect the normalized performance metrics, $x_i(k)$ can be expressed in terms of its phase value $\phi_i(k)$ as

$$x_i(k) = e^{j\phi_i(k)}. \quad (2)$$

The $\phi_i(k)$ is in the set $\mathbb{R}$, and $\mathbb{R}$ denotes the real number space.

The auto-correlation function of the waveform $\mathbf{x}_i$ can be expressed as

$$r_{ii}(\tau) = \sum_{k=0}^{N-1} x_i^*(k) x_i(k+\tau), \tau \in [-N+1, N-1] \quad (3)$$

where $x_i^*(k)$ is the complex conjugate of the $(k+1)$th element.

Similarly, the cross-correlation function of any two waveforms $\mathbf{x}_i$ and $\mathbf{x}_j$ can be expressed as

$$r_{ij}(\tau) = \sum_{k=0}^{N-1} x_i^*(k) x_j(k+\tau), \tau \in [-N+1, N-1] \quad (4)$$

where $x_j(k+\tau)$ is the $(k+\tau+1)$th element of the jth waveform.

PSL and ISL normalized by waveform energy are widely recognized as performance metrics for orthogonal waveform set design. According to (2), the waveform energy under constant modulus constraint is N^2. PSL is defined as

$$\mathrm{PSL} \triangleq \frac{1}{N^2} \max\left(\max_{\tau \neq 0, i=j} \left| r_{ij}(\tau) \right|^2, \max_{i \neq j} \left| r_{ij}(\tau) \right|^2 \right). \quad (5)$$

ISL is defined as

$$\mathrm{ISL} \triangleq \frac{1}{N^2} \left(\sum_{\tau \neq 0, i=j} \left| r_{ij}(\tau) \right|^2 + \sum_{i \neq j} \left| r_{ij}(\tau) \right|^2 \right). \quad (6)$$

Therefore, the PSL optimization can be modeled as

$$\begin{aligned} \min_{\{\mathbf{x}_i\}} \quad & \mathrm{PSL} \\ \text{s.t.} \quad & x_i(k) = e^{j\phi_i(k)}, \\ & \phi_i(k) \in (-\infty, +\infty), \\ & 0 \le k \le N-1, \\ & 1 \le i \le M. \end{aligned} \quad (7)$$

This is a nonconvex and NP-hard minimax optimization problem. The representation of the complex exponential alleviates the complexity of optimization by converting the constant modulus constraint into an unconstrained one.

Since the objective function of problem (7) is nondifferentiable, it cannot be solved directly by GD algorithm. To obtain a differentiable objective function, we introduce the LSE function mentioned in the literature [51], [52], [53] to approximate PSL metric. The LSE_t function is defined as

$$\mathrm{LSE}_t(\mathbf{z}) = t \log\left(\sum_{i=1}^{n} e^{\frac{z_i}{t}} \right) \quad (8)$$

where $\mathbf{z} = [z_1, z_2, \ldots, z_n]^{\mathrm{T}}$ is an n-dimensional column vector and the subscript t denotes the degree of approximation. The LSE_t function has a boundary relationship with the max function [53]

$$\max(z_1, \ldots, z_n) \le \mathrm{LSE}_t(\mathbf{z}) \le \max(z_1, \ldots, z_n) + t \log(n). \quad (9)$$

The first equality sign holds at $n = 1$ and the second at $z_1 = z_2 = \cdots = z_n$. From the squeeze theorem, we can obtain

$$\lim_{t \to 0} \mathrm{LSE}_t(\mathbf{z}) = \max(z_1, \ldots, z_n). \quad (10)$$

So we can use LSE_t to approximate PSL. The term $t \log(n)$ in (9) indicates that the fewer the number of variables, the better the degree of approximation. In addition, the LSE_t can be calculated as follows to avoid exponential overflow:

$$\mathrm{LSE}_t(\mathbf{z}) = \max(\mathbf{z}) + t \log\left(\sum_{i=1}^{n} e^{\frac{z_i - \max(\mathbf{z})}{t}} \right). \quad (11)$$

Consequently, the unconstrained optimization problem can be reformulated based on the smooth approximation of the objective function as follows:

$$\begin{aligned} \min_{\{\mathbf{x}_i\}} \quad & \mathrm{LSE}_t(\mathbf{p}) \\ \text{s.t.} \quad & x_i(k) = e^{j\phi_i(k)} \\ & \phi_i(k) \in (-\infty, +\infty) \\ & 0 \le k \le N-1 \\ & 1 \le i \le M \end{aligned} \quad (12)$$

where $\mathbf{p}$ is the column vector consisting of all the values of the normalized correlation function and satisfying that all combinations of (i, j, τ) are in the set $\{(i, j, \tau) | i \neq j \ or \ \tau \neq 0\}$.

Also, a more general weighted problem can be formulated as follows:

$$\begin{aligned} \min_{\{\mathbf{x}_i\}} \quad & \mathrm{LSE}_t(\mathbf{p}_w) \\ \text{s.t.} \quad & x_i(k) = e^{j\phi_i(k)} \\ & \phi_i(k) \in (-\infty, +\infty) \\ & 0 \le k \le N-1 \\ & 1 \le i \le M \end{aligned} \quad (13)$$

where $\mathbf{p}_w$ is the vector consisting of all the values of the weighted normalized correlation function and satisfying

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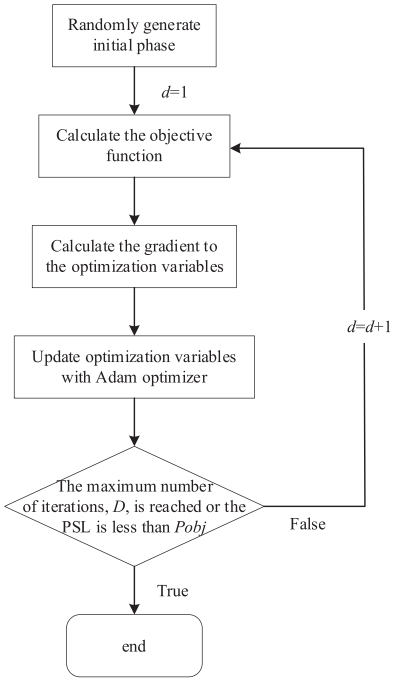

Fig. 1. Flowchart of the GD-based algorithm.

that all combinations of (i, j, τ) are in the set $\{(i, j, \tau)|i \neq j$ or $\tau \neq 0\}$.

The weighted normalized correlation function is defined as

$$r_{ij}^{\text{weighted}}(\tau) = w_{ij}(\tau)\, r_{ij}(\tau) \tag{14}$$

where $w_{ij}(\tau)$ is the weight correspond to each normalized correlation function, and satisfies $w_{ij}(\tau) \in [0, 1]$. For the conjugate symmetry of the correlation function, $w_{ij}(\tau)$ should always equal to $w_{ji}(-\tau)$. If we choose $w_{ij}(\tau) = 1$ for all combinations of (i, j, τ), then problem (13) reduces to (12).

III. THE GD-BASED ALGORITHM

Solving optimization problem (7) directly is hard for existing optimizers due to the nondifferentiability of its objective function. However, by approximating PSL using the LSE function, the optimization problem (13) becomes easy to solve because it is differentiable and its feasible region remains unconstrained. Therefore, we propose a GD-based algorithm that employs the Adam optimizer to solve the approximate problem (13). The flowchart of the GD-based algorithm is shown in Fig. 1.

GD is an efficient iterative optimization method that gradually converges to the optimal or local optimal solution based on the gradient of the objective function. The analytic gradient of high-dimensional functions like in optimization problem (13) is usually complex. However, we extended the gradient in [27] to derive the analytic gradient of optimization problem (13). This section presents the analytical gradient and the way to calculate the objective function of approximation problem (13), along with specific iterative steps for solving the problem by employing the Adam optimization algorithm.

A. Analytic Gradient of the Objective Function

According to [27], phase gradients of auto-correlation and cross-correlation can be extended to MIMO systems. The gradient of the objective function with respect to the phases can be expressed as

$$\nabla_{\phi_u} \text{LSE}_t = \Im(\mathbf{x}_u^* \circ (\nabla_{\mathbf{x}_u} \text{LSE}_t)) \tag{15}$$

where $\nabla_{\mathbf{x}_u} \text{LSE}_t$ is the gradient of the objective function with respect to the waveforms. Applying the chain rule enables a straightforward computation of $\nabla_{\mathbf{x}_u} \text{LSE}_t$. The elementwise expression of $\nabla_{\mathbf{x}_u} \text{LSE}_t$ is

$$\frac{\partial \text{LSE}_t}{\partial x_u(v)} = \sum_{i=1}^{M} \sum_{j=1}^{M} \sum_{\tau=-N+1}^{N-1} w_{ij}(\tau) \frac{\partial \text{LSE}_t}{\partial \left|r_{ij}(\tau)\right|^2} \frac{\partial \left|r_{ij}(\tau)\right|^2}{\partial x_u(v)} \tag{16}$$

where $x_u(v)$ denotes the $(v+1)$th element in the uth sequence.

Each term of (16) consists of two parts. The first part is the partial derivative of the LSE_t function with respect to $|r_{ij}(\tau)|^2$. According to (8), we can get

$$\frac{\partial \text{LSE}_t}{\partial \left|r_{ij}(\tau)\right|^2} = \frac{1}{N^2} e^{(\left|r_{ij}(\tau)\right|^2/N^2 - \text{LSE}_t)/t} \tag{17}$$

where $e^{\text{LSE}_t/t}$ denotes the summation part within the logarithmic term of LSE_t. According to (9), the power of the exponential term $(|r_{ij}(\tau)|^2/N^2 - \text{LSE}_t)/t$ is in the interval $[-\log(n), \log(n)]$, then the term $e^{(|r_{ij}(\tau)|^2/N^2 - \text{LSE}_t)/t}$ will not overflow during calculation. The second part of (16) is the partial derivative of $|r_{ij}(\tau)|^2$ with respect to $x_u(v)$, which is nonzero only when $u = i, j$.

We give $\nabla_{\mathbf{x}_u} \text{LSE}_t$ directly here, and the detailed derivation is in Appendix-A. Define a $2N-1$ dimensional vector $\mathbf{g}_{ui}$ elementwise as

$$g_{ui}(\tau) = w_{ui}(\tau) \frac{\partial \text{LSE}_t}{\partial |r_{ui}(\tau)|^2} r_{ui}(\tau) \tag{18}$$

where $\tau \in [-N+1, N-1]$. Then, a $3N-2$ dimensional vector $\mathbf{h}_{ui}$ can be obtained by correlating $\mathbf{g}_{ui}$ and $\mathbf{x}_u$

$$h_{ui}(v) = \sum_{\tau=-N+1}^{N-1} g_{ui}^*(\tau) x_u(\tau + v). \tag{19}$$

Therefore, the gradient of the objective function with respect to the waveforms can be expressed as

$$\nabla_{\mathbf{x}_u} \text{LSE}_t = 2 \sum_{i=1}^{M} \mathbf{h}_{ui}(\tau : \tau + N) \tag{20}$$

where $\mathbf{h}_{ui}(\tau : \tau + N)$ represents the vector that consisting of the τth to $(\tau + N)$th elements of $\mathbf{h}_{ui}$.

According to (15)–(20), the gradient of the objective function to the phases can be obtained. The gradient calculation involves two correlation operations, which can be accelerated by FFT. One computes the correlation function

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of the waveform set, while the other computes the correlation function of $\mathbf{g}_{ui}$ and $\mathbf{x}_u$.

B. Calculation of the Objective Function

The expression of the gradient involves the objective function, thus this section presents the method to calculate the objective function of the problem (13). First, arrange the phases of all MN elements in M waveforms into an $N \times M$ dimensional phase matrix

$$\mathbf{\Phi} = \begin{bmatrix} \phi_1(0) & \phi_2(0) & \cdots & \phi_M(0) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_1(N-1) & \phi_2(N-1) & \cdots & \phi_M(N-1) \end{bmatrix} \in \mathbb{R}^{N\times M}. \tag{21}$$

Then, according to (2), the waveform matrix can be calculated element-by-element from the phase matrix as follows:

$$\begin{aligned} \mathbf{X} &= \begin{bmatrix} x_1(0) & x_2(0) & \cdots & x_M(0) \\ \vdots & \vdots & \ddots & \vdots \\ x_1(N-1) & x_2(N-1) & \cdots & x_M(N-1) \end{bmatrix} \\ &= [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_i, \ldots, \mathbf{x}_M] \in \mathbb{C}^{N\times M} \end{aligned} \tag{22}$$

where $\mathbf{x}_i$ denotes the ith waveform, which is consistent with the definition of (1).

According to the conjugate symmetry of the correlation function, the calculation of the LSE_t function requires only half of the correlation results. First, take the conjugate of the waveform matrix $\mathbf{X}$ and pad $N-1$ rows with zeros

$$\begin{aligned} \tilde{\mathbf{X}} &= \begin{bmatrix} \mathbf{0}_{(N-1)\times M} \\ \mathbf{X}^* \end{bmatrix} \\ &= [\tilde{\mathbf{x}}_1, \ldots, \tilde{\mathbf{x}}_i, \ldots, \tilde{\mathbf{x}}_M] \end{aligned} \tag{23}$$

where $\mathbf{X}^*$ is the conjugate matrix of $\mathbf{X}$. The row transformation matrix can be defined as a square matrix with all antidiagonal elements equal to 1 and all other elements equal to 0

$$\mathbf{P} = \begin{bmatrix} 0 & 0 & \ldots & 0 & 1 \\ 0 & 0 & \ldots & 1 & 0 \\ \vdots & \vdots & .. & \vdots & \vdots \\ 0 & 1 & \ldots & 0 & 0 \\ 1 & 0 & \ldots & 0 & 0 \end{bmatrix}. \tag{24}$$

The flipped waveform matrix $\mathbf{X}^{\mathrm{f}}$ can be obtained by left-multiplying the row transformation matrix $\mathbf{P}$, i.e.,

$$\begin{aligned} \mathbf{X}^{\mathrm{f}} &= \mathbf{PX} \\ &= [\mathbf{x}_1^{\mathrm{f}}, \ldots, \mathbf{x}_i^{\mathrm{f}}, \ldots, \mathbf{x}_M^{\mathrm{f}}] \end{aligned} \tag{25}$$

where $\mathbf{x}_i^{\mathrm{f}} = [x_i(N-1), \ldots, x_i(0)]^{\mathrm{T}}$ is the ith waveform after flipping. Then, the nonzero padding convolution of all columns $\tilde{\mathbf{x}}_i (1 \le i \le M)$ of $\tilde{\mathbf{X}}$ and all columns $\mathbf{x}_j^{\mathrm{f}} (1 \le j \le M)$ of $\mathbf{X}^{\mathrm{f}}$ is calculated and rearranged into a $N \times M^2$ dimensional matrix

$$\mathbf{R} = [\tilde{\mathbf{x}}_1 \otimes \mathbf{x}_1^{\mathrm{f}}, \ldots, \tilde{\mathbf{x}}_1 \otimes \mathbf{x}_M^{\mathrm{f}}, \ldots, \tilde{\mathbf{x}}_M \otimes \mathbf{x}_M^{\mathrm{f}}] \tag{26}$$

where $\otimes$ represents the convolution operation without zero padding. It is worth noting that FFT can be used to speed up convolution operation. The flipped correlation function is defined as $\mathbf{r}_{ij}^{\mathrm{f}} = [r_{ij}(N-1), \ldots, r_{ij}(0)]^{\mathrm{T}}$, then the matrix $\mathbf{R}$ can be expressed as

$$\mathbf{R} = \left[\mathbf{r}_{11}^{\mathrm{f}}, \ldots, \mathbf{r}_{1M}^{\mathrm{f}}, \ldots, \mathbf{r}_{M1}^{\mathrm{f}}, \ldots, \mathbf{r}_{MM}^{\mathrm{f}}\right]. \tag{27}$$

The matrix $\mathbf{R}$ contains all auto-correlations and cross-correlations, but the objective function is not related to the mainlobe of the auto-correlation. Therefore, it is necessary to construct an indicator matrix that eliminates the mainlobe of the auto-correlation in the matrix $\mathbf{R}$

$$\mathbf{e}_i = [\overbrace{\underbrace{0, \ldots, 1}_{i}, \ldots, 0}^{N}]^{\mathrm{T}}. \tag{28}$$

The $\mathbf{e}_i$ denotes an N-dimensional unit vector whose elements are all zeros except the ith element being one. Using M^2 unit vector $\mathbf{e}_i$ to mark the position where the mainlobe of the auto-correlation in the matrix $\mathbf{R}$. Define an $N \times M^2$ dimensional indicator matrix $\mathbf{J}$ consisting of $\mathbf{e}_i$ and $\mathbf{0}_{N\times 1}$ as

$$\mathbf{J} = [\underbrace{\mathbf{e}_N, \ldots, \mathbf{0}_{N\times 1}}_{M}, \ldots, \underbrace{\mathbf{0}_{N\times 1}, \ldots, \mathbf{e}_N}_{M}]. \tag{29}$$

The element 1 in the matrix $\mathbf{J}$ indicates that the corresponding position of the correlation matrix $\mathbf{R}$ is the mainlobe of the auto-correlation. The normalized correlation matrix $\tilde{\mathbf{R}}$, with the auto-correlation mainlobe setting to zero, can be obtained by elementwise multiplication of the logically inverted indicator matrix $\mathbf{J}$ and the correlation matrix $\mathbf{R}$, i.e.,

$$\tilde{\mathbf{R}} = \frac{1}{N}(\neg \mathbf{J}) \circ \mathbf{R} \tag{30}$$

where $\neg$ denotes the element-by-element logical inverse of a matrix. Finally, take the elementwise modulus square of the normalized correlation matrix $\tilde{\mathbf{R}}$, and the objective function of the optimization problem (12) can be written as

$$\mathrm{LSE}_t(\mathbf{p}) = \mathrm{LSE}_t(\mathrm{vec}\{||\tilde{\mathbf{R}}||^2\}) \tag{31}$$

where $\mathrm{vec}\{||\tilde{\mathbf{R}}||^2\}$ is an $NM^2 \times 1$ dimensional vector.

For problem (13), just construct another indicator matrix $\mathbf{J}_w$ as

$$\mathbf{J}_w = [\underbrace{\mathbf{w}_{11}, \ldots, \mathbf{w}_{1M}}_{M}, \ldots, \underbrace{\mathbf{w}_{M1}, \ldots, \mathbf{w}_{MM}}_{M}] \tag{32}$$

where $\mathbf{w}_{ij} = [w_{ij}(N-1), \ldots, w_{ij}(0)]^{\mathrm{T}}$, and $w_{ij}(\tau)$ is the weight corresponding to each normalized correlation function.

The weighted correlation matrix $\tilde{\mathbf{R}}_w$ can be obtained by elementwise multiplication of the new indicator matrix $\mathbf{J}_w$, i.e.,

$$\tilde{\mathbf{R}}_w = \mathbf{J}_w \circ \tilde{\mathbf{R}}. \tag{33}$$

By squaring the elementwise modulus of the normalized correlation matrix $\tilde{\mathbf{R}}_w$, the objective function of the

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optimization problem (13) can be written as

$$\mathrm{LSE}_t(\mathbf{p}_w) = \mathrm{LSE}_t(\mathrm{vec}\{||\tilde{\mathbf{R}}_w||^2\}) \tag{34}$$

where $\mathrm{vec}\{||\tilde{\mathbf{R}}_w||^2\}$ is also an $NM^2 \times 1$ dimensional vector.

C. Parameter Update

The gradient of the objective function $\nabla_\phi \mathrm{LSE}_t$ is calculated by (15)–(20). The optimization problem (13) is then tackled by employing iterative steps in the direction of GD. The choice of optimizer significantly affects the convergence rate and solution quality. Traditional first-order GD algorithms tend to get stuck in a local optima and struggle to escape saddle points. Hence, momentum-based optimizers have been widely adopted to address these issues. This strategy resembles a massive particle moving on the surface of an objective function with friction. When encountering local optima or saddle points, the particle maintains a nonzero velocity, allowing it to advance and converge toward higher quality solutions. In this article, we use the Adam optimizer [54], which combines momentum and root mean squared propagation. Such optimizers are widely used in neural networks. Its key feature involves estimating the first and second moments of gradients and updating each parameter with independent adaptive learning rates. The Adam optimizer exhibits high computational efficiency and can be easily implemented. The update process of the Adam optimizer for the phase $\phi^{(d)}$ is given as follows:

$$\begin{aligned}
m_\phi^{(d+1)} &= \beta_1 m_\phi^{(d)} + (1-\beta_1)\nabla_\phi \mathrm{LSE}_t^{(d)},\\
v_\phi^{(d+1)} &= \beta_2 v_\phi^{(d)} + (1-\beta_2)\left(\nabla_\phi \mathrm{LSE}_t^{(d)}\right)^2,\\
\hat{m}_\phi &= \frac{m_\phi^{(d+1)}}{1-\beta_1^d},\\
\hat{v}_\phi &= \frac{v_\phi^{(d+1)}}{1-\beta_2^d},\\
\phi^{(d+1)} &= \phi^{(d)} - \gamma \frac{\hat{m}_\phi}{\sqrt{\hat{v}_\phi}+\epsilon}.
\end{aligned} \tag{35}$$

The superscript d in parentheses denotes the number of iterations. $m_\phi^{(d+1)}$ and $v_\phi^{(d+1)}$ denote the biased first-order and second-order moment estimation of the gradient. β_1 and β_2 are forgetting factors. ${\beta_1}^d$ and ${\beta_2}^d$ are, respectively, the dth powers of β_1 and β_2. $\hat{m}_\phi$ and $\hat{v}_\phi$ are the bias-corrected first-order and second-order moment estimation, respectively. γ is the step size under current direction of gradient estimation, and ϵ is a small scalar to avoid the denominator being zero. Typically, β_1, β_2, and ϵ are set to be 0.9, 0.999, and 10^{-8}, respectively.

D. Summary of the GD-Based Algorithm

The GD-based algorithm can be summarized as Algorithm 1. First, given the number of waveforms M, the sequence length N, the desired PSL metric $Pobj$, maximum number of iterations D, and the exponential step decay function $h(\gamma)$ of Adam algorithm. Then, initialize the step size $\gamma^{(0)}$, and phase $\phi_i^{(0)}(k)$, which will be optimized. The objective function and gradient are computed in a loop to update the optimization variables. Finally, the algorithm terminates when the maximum number of iterations, D, is reached or the PSL is less than $Pobj$. At this time, an approximate orthogonal phase-coded waveform set is obtained. The step size of each iteration is updated by the decay function $h(\gamma)$.

Algorithm 1: GD-Based Algorithm.

Initialization: $\phi_i^{(0)}(k)$, $1 \le i \le M$, $0 \le k \le N-1$, $\gamma^{(0)}$.
Construct $\mathbf{J}$ and $\mathbf{J}_w$ according to (28)–(29) and (32)
while $d \le D$ **do**
  Compute $\mathbf{X}$ from $\mathbf{\Phi}$
  Compute $\mathrm{LSE}_t(\mathrm{vec}\{||\tilde{\mathbf{R}}_w||^2\})$
  if $\max(\mathrm{vec}\{||\tilde{\mathbf{R}}||^2\}) < Pobj$
    break
  end if
  Compute the gradient according to (15)–(20)
  Update $\mathbf{\Phi}$ according to (35)
  $\gamma^{(d)} \leftarrow h(\gamma^{(d-1)})$
  $d \leftarrow d+1$
end while
$x_i(k) \leftarrow x_i^{(d)}(k)$, $1 \le i \le M$, $0 \le k \le N-1$

The initialization of the phase $\phi_i(k)$ affects the convergence performance of the algorithm. If the phase $\phi_i(k)$ are initialized with the same value, the resulting $x_i(k)$ are also identical. Substituting all identical $x_i(k)$ into (15)–(20), we can see that

$$\frac{\partial \mathrm{LSE}_t}{\partial \phi_u(v)} = 0. \tag{36}$$

The objective function currently is located at a maximum value point, and the derivatives in all directions are zero. Even when using the momentum GD algorithm, it is not possible to cross over a local maximum, as the first iteration reaches a local maximum. Therefore, a normal distribution with mean 0 rad and standard deviation 0.01 rad is used in the following to initialize the phase.

IV. LINKS TO NEURAL NETWORKS

This section discusses connections between the GD-based algorithm and neural networks in the context of phase-coded waveform design. By denoting the algorithm as a neural network-like backpropagation structure, it is straightforward to leverage mature software frameworks to solve the waveform design problem (13).

A simple fully connected neural network is shown in Fig. 2. This network is composed of only an input layer and an output layer, and has no hidden layer. The relation between the output $\mathbf{y}_{NN}$ and input $\mathbf{x}_{NN}$ of the network can be expressed as

$$\mathbf{y}_{NN} = g(\mathbf{W}\mathbf{x}_{NN} + \mathbf{b}) \in \mathbb{R}^{NM\times 1} \tag{37}$$

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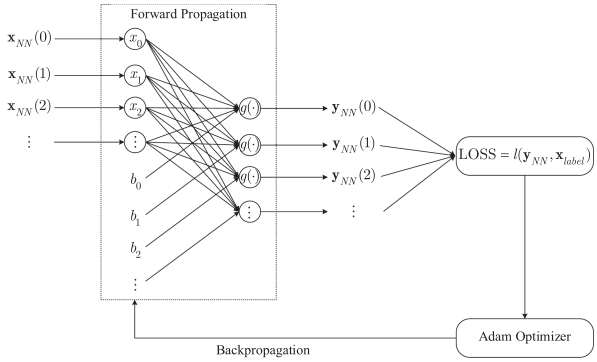

Fig. 2. Simple fully connected neural network.

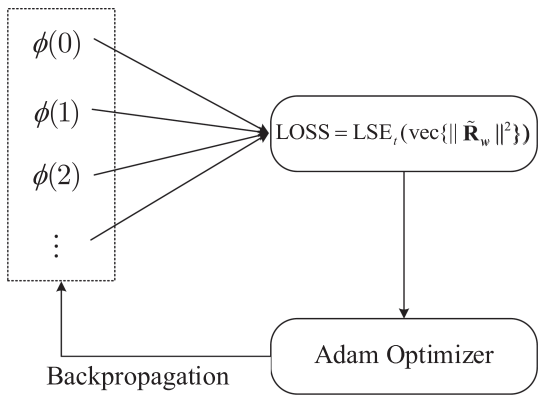

Fig. 3. Backpropagation structure of the optimization problem (13).

where $\mathbf{y}_{NN}$, $\mathbf{x}_{NN}$, and $\mathbf{b}$ are all $NM \times 1$ dimensional real vectors. The upper case $\mathbf{W}$ is an $NM \times NM$ dimensional real matrix and different from the lower case $\mathbf{w}_{ij}$ mentioned before. The activation function is $g(\cdot)$. The loss function can be calculated from the network output and labels as

$$\text{LOSS} = l(\mathbf{y}_{NN}, \mathbf{x}_{\text{label}}) \tag{38}$$

where $l(\cdot)$ denotes the detailed form of the loss function.

For waveform design problems, there is no equivalent data-label pair as typically found in neural network applications. This allows for further simplification of the network structure. Let $\mathbf{W}$ be a zero matrix and $g(\cdot)$ be the constant activation function, then the network output $\mathbf{y}_{NN}$ is only related to $\mathbf{b}$. Replace $\mathbf{b}$ with vec$\{\mathbf{\Phi}\}$, the network output can be expressed as

$$\mathbf{y}_{NN} = \text{vec}\{\mathbf{\Phi}\} \in \mathbb{R}^{NM\times 1} \tag{39}$$

where $\mathbf{y}_{NN}$ is another representation of the phase. The loss function independent of the label can be expressed as

$$\text{LOSS} = \text{LSE}_t(\text{vec}\{||\tilde{\mathbf{R}}_w||^2\}) \tag{40}$$

where $\text{LSE}_t(\text{vec}\{||\hat{\mathbf{R}}_w||^2\})$ can be computed by the same process according to (22)–(34). Thus, the optimization problem (13) can be represented using the backpropagation architecture illustrated in Fig. 3.

It can be observed that while Fig. 3 represents a very basic neural network, it encompasses the entire process of solving the optimization problem (13) using the GD method. Based on the structure shown in Fig. 3, we can employ mature software frameworks in the field of deep learning, such as PyTorch [51], TensorFlow [52], Caffe [53], etc., to tackle the waveform design problem. These frameworks can differentiate functions automatically, greatly reducing the complexity of gradient computation. According to Griewank's analysis of the complexity of automatic differentiation [61], the computation of gradients can be at most five times the computational cost of the function itself, regardless of the dimensionality of the variables.

Algorithm 2: GD-Based Algorithm With Neural Network.

Initialization: $\phi_i^{(0)}(k)$, $1 \le i \le M$, $0 \le k \le N-1$, $\gamma^{(0)}$.
Construct $\mathbf{J}$ and $\mathbf{J}_w$ according to (28)–(29) and (32)
while $d \le D$ **do**
  Compute $\mathbf{X}$ from $\mathbf{\Phi}$
  Compute $\text{LSE}_t(\text{vec}\{||\tilde{\mathbf{R}}_w||^2\})$
  if $\max(\text{vec}\{||\tilde{\mathbf{R}}||^2\}) < Pobj$
    break
  end if
  Automatic differentiation
  Update $\mathbf{\Phi}$ according to (35)
  $\gamma^{(d)} \leftarrow h(\gamma^{(d-1)})$
  $d \leftarrow d+1$
end while
$x_i(k) \leftarrow x_i^{(d)}(k)$, $1 \le i \le M$, $0 \le k \le N-1$

Mature software frameworks, like PyTorch, include commonly used loss functions, such as the quadratic loss function, cross-entropy loss function, and log-likelihood loss function. Although these functions cannot be directly applied to solve waveform design problems due to their reliance on network outputs and labels as input parameters, a few minor modifications can eliminate this dependency. In particular, we initially inherited the loss function class, which computed loss from both network outputs and labels within its forward function. Subsequently, a refined forward function within the loss function class was implemented to accommodate the waveform design problem by ignoring the data labels, eliminating the requirement for labels in the loss calculation. The GD-based algorithm implemented by neural network can be summarized as Algorithm 2. The main difference from Algorithm 1 lies in the calculation method of gradients at the ninth step.

The use of neural network framework greatly reduces the programming effort and computation time, and obtains the waveform sets with the same performance as that achieved by analytic GD Algorithm 1. At the same time, GPUs can accelerate most vector operations and parallel operations, further speeding up waveform design.

V. NUMERICAL RESULTS

This section presents numerical examples to illustrate the performance of the GD-based algorithm under different waveform set parameters. We will start with demonstrating the equivalence between the GD-based Algorithm 1

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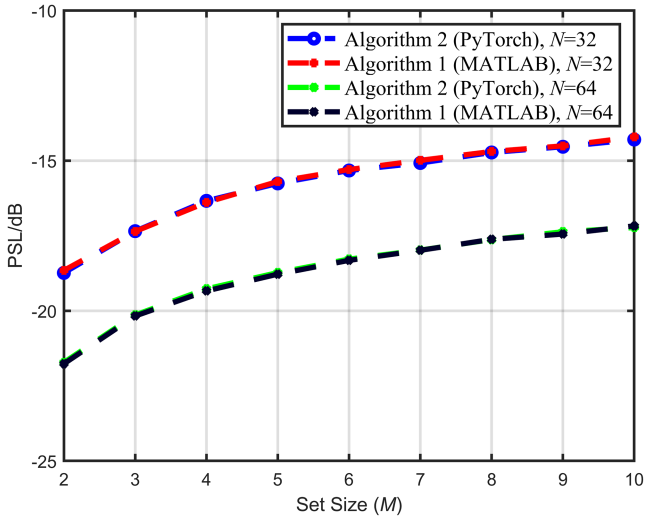

Fig. 4. PSL obtained by two different implementations of GD-based algorithm under $N = \{32,64\}$.

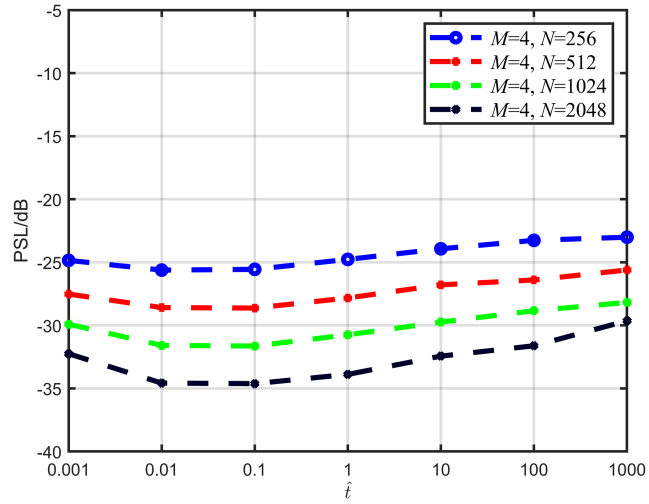

Fig. 5. PSL obtained by the GD-based algorithm with different $\hat{t}$.

in Section III and Algorithm 2 in Section IV, which is implemented by neural network. Then, the comparison with state-of-the-art methods, including Multi-CAN [22], ISL-New [26], MM-Corr [25], Primal-Dual [28], p-MM [29], WeBEST-v [30], and CON [46], is discussed based on the neural network implementation. All experiments are performed on a computer equipped with an i7-13700K CPU, RTX4090 GPU, and 64GB RAM. The neural network used by Algorithm 2 is constructed by PyTorch 2.0 backend in Python 3.8.

The normalized Welch bound are used for performance comparison. The PSL bound can be expressed as [22], [25], [28], and [29]

$$\mathrm{PSL}_{\mathrm{welch}} = \frac{M-1}{2MN-M-1} \tag{41}$$

and the ISL bound can be expressed as [22], [26], [29]

$$\mathrm{ISL}_{\mathrm{welch}} = M(M-1) \tag{42}$$

where M is the set size and N is the sequence length. All the obtained PSL and ISL values are expressed in decibels (dB).

A. Effectiveness of the Backpropagation Structure

In this subsection, we compare the optimization results obtained by the backpropagation structure Algorithm 2 implemented by PyTorch and Algorithm 1 implemented by MATLAB. The phase is initialized by a normal distribution with mean 0 rad and standard deviation 0.01 rad. The learning rate of Adam optimizer starts at 0.01, number of iterations D is 20 000, and the exponential step decay function is $h(\gamma) = 0.9999\gamma$. The approximation parameter t is 0.0004. We choose the weights $w_{ij}(\tau) = 1$ for all combinations of (i, j, τ) to suppress all correlation sidelobes.

Fig. 4 illustrates a comparison of PSL obtained by two different implementations of the GD-based algorithm as $M = \{2,3,4,5,6,7,8,9,10\}$ and $N = \{32,64\}$. The PSLs obtained from the two implementations show a negligible difference of up to 0.09 dB for different sequence lengths. This observation highlights the efficacy of utilizing neural networks with backpropagation structure to implement GD-based algorithms. The advantage of using the neural network software framework is that it greatly simplifies programming. Generally, only a few parameters need to be configured for waveform sets design tasks.

Since Algorithm 2 can achieve almost the same performance with Algorithm 1 and have faster optimization speed, we will use the results of Algorithm 2 for analysis and comparison in the following subsections. And the computational time will be discussed in the last subsection.

B. Approximation Parameter t

This subsection analyzes the influence of approximation parameter t on the optimization performance. It can be seen that the LSE_t tends to PSL as t tends to zero from (10). However, as t approaches zero, the smoothness of LSE_t function decreases, causing gradients to change dramatically. This necessitates smaller iteration step sizes and longer iterations during optimization. Conversely, when t is large, the LSE_t function exhibits a significant difference from the max function, also leading to a decrease in optimization performance. Therefore, it is important to select an appropriate t that balances the optimization efficiency with the approximation to achieve better performance.

When aiming to achieve better approximation, it is advisable to select a smaller t in the case of long sequence. That is, different sequence lengths correspond to different t. For this reason, we analyze PSL with different $\hat{t} = tMN$, to avoid the influence of M and N on the choice of t. The simulation parameters, such as the initial value of the phase, learning rate, and weights, are the same as those in the previous subsection. Fig. 5 shows the relationship between the PSL obtained by the GD-based algorithm and $\hat{t}$ in the set $\{0.001,0.01,0.1,1.10,100,1000\}$. When $\hat{t} = 0.001$, the PSL of different set size under the same number of iterations is relatively high. After $\hat{t} \geq 0.01$, as the smoothness of LSE_t

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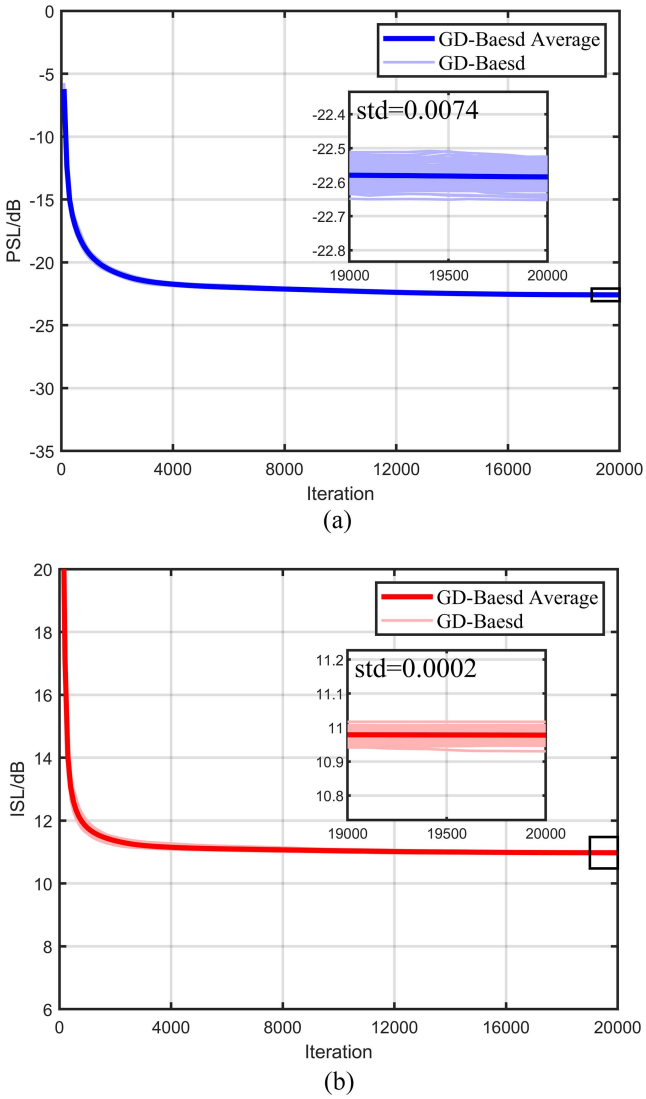

Fig. 6. Convergence curves of PSL and ISL obtained by the GD-based algorithm over 100 independent trials under $M = 4$ and $N = 128$. (a) PSL. (b) ISL.

function increases, PSL gradually decreases (better). However, the PSL performance degrades with the increase of $\hat{t}$ after $\hat{t} \geq 1$ due to the reduced approximation. To achieve the best PSL performance, $\hat{t}$ is set to 0.01 in the following subsections. That is, $t = 0.01/MN$.

C. Stability

This subsection examines the stability of the GD-based algorithm. The simulation parameters remain consistent with the previous subsection. Fig. 6 illustrates the convergence curves of PSL and ISL over 100 independent trials. The light-colored lines represent individual convergence curves, and the dark-colored lines represent the average of all convergence curves.

In Fig. 6, ISL exhibits a similar monotonic decreasing trend as PSL. PSL is more sensitive to variations of the phase, resulting in a broader region of light shadowing at the same scale. The standard deviations of PSL and ISL obtained from repeated experiments are both below 0.01 dB, although the standard deviation of PSL is slightly higher than ISL. Moreover, the results are nearly the same from different initial phase-coded waveforms, indicating the favorable stability of the GD-based algorithm.

D. Comparison With the Other Methods

This subsection compares the PSL and ISL performance of the GD-based algorithm under different waveform set sizes with the state-of-the-art phase-coded waveform design algorithms, such as multi-CAN [22], ISL-New [26], MM-Corr [25], Primal–Dual [28], p-MM [29], WeBEST-v [30], and CON [46]. Multi-CAN, ISL-New, and MM-Corr optimize ISL, while Primal–Dual and the p-MM algorithm optimize PSL. CON optimizes a weighted combination of PSL and ISL. WeBEST-v optimizes PSL or ISL under different values of p. For the GD-based algorithm, the phase is initialized by a normal distribution with mean of 0 rad and standard deviation of 0.01 rad. The learning rate of Adam optimizer starts at 0.01, number of iterations D is 20 000, and the exponential step decay function is $h(\gamma) = 0.9999\gamma$. We choose the weights $w_{ij}(\tau) = 1$ for all combinations of (i, j, τ) to suppress all correlation sidelobes. According to [46], the best-performing CON is a neural network with 10 residual blocks. The learning rate of CON is set to 0.0005, and the number of iterations D remains 20 000. The loss weights are selected as $l_1 = l_2 = l_3 = 1$, $l_4 = l_5 = 0.5$. To get better PSL performance, the parameter p of WeBEST-v is set to 10 [30]. Besides, the stop condition, backtracking line search parameters, and maximum number of iterations of WeBEST-v are selected as $\zeta \leq 10^{-9}$, $\alpha = 0.4$, $\beta = 0.8$, 10^5, respectively.

Fig. 7(a) illustrates a comparison of PSL performance between the GD-based algorithm and other algorithms under $M = 4$ and N in the set $\{32,64,128,256,512,1024\}$. The PSL obtained by all algorithms gradually decreases as N increases, in line with the trend of the lower bound on PSL provided by Welch [20]. The lower bound is indicated in the graph by an unmarked black dashed line. The PSL values obtained by multi-CAN, ISL-New, and MM-Corr are nearly identical. Both p-MM and Primal–Dual directly optimize PSL, resulting in PSL value is 3–5 dB better than ISL minimization algorithms for various set sizes. The PSL obtained by CON is lower by 1–2 dB than p-MM and Primal–Dual for smaller set sizes. But with the increase of sequence length, the PSL obtained by those three algorithms gets closer. The PSL obtained by WeBEST-v is the lowest among the state-of-the-art algorithms.

Despite this, a gap of around 5 dB from the PSL lower bound still exists for the abovementioned algorithms. The proposed GD-based algorithm consistently outperforms the existing algorithms and achieves better PSL value that is 1–2 dB lower than state-of-the-art algorithms for larger sequence length. For instance, in the case of $M = 4$ and $N = 1024$, the PSL obtained by the GD-based algorithm is 1.5 dB lower than the WeBEST-v algorithm.

Although the proposed GD-based algorithm focuses on PSL optimization, it can achieve good ISL performance as well. In Fig. 7(b), a comparison of ISL under the same set

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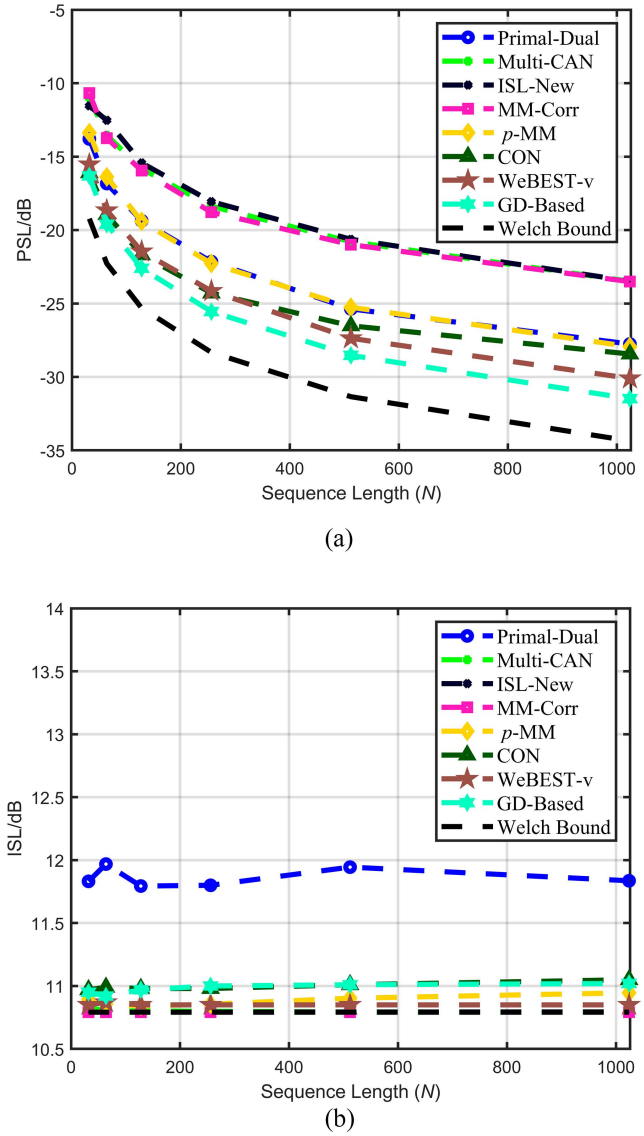

Fig. 7. Metrics values obtained by the GD-Based algorithm under $M = 4$. (a) PSL. (b) ISL.

sizes as Fig. 7(a) between the GD-based algorithm and other algorithms is illustrated. When M is constant, the ISL exhibits minimal variation with changing the sequence length, consistent with the lower bound on ISL provided by Welch [20]. The ISL obtained by Primal–Dual is the highest among all algorithms, about 1 dB above the Welch lower bound. Multi-CAN, ISL-New, and MM-Corr all nearly reach the Welch lower bound within 0.3 dB. Among the algorithms optimizing PSL, p-MM achieves a lower ISL, 0.2 dB above the Welch lower bound, compared to Primal–Dual, while the GD-based algorithm yields 0.05 dB higher ISL than p-MM but still significantly lower than that obtained by Primal–Dual. The ISL obtained by CON is nearly identical to that obtained by the GD-based algorithm, while the ISL obtained by WeBEST-v is lower than that obtained by the GD-based algorithm by about 0.1 dB.

Considering both PSL and ISL, under $M = 4$, the proposed algorithm can achieve the best PSL suppression among all the methods, with a clear advantage of up to

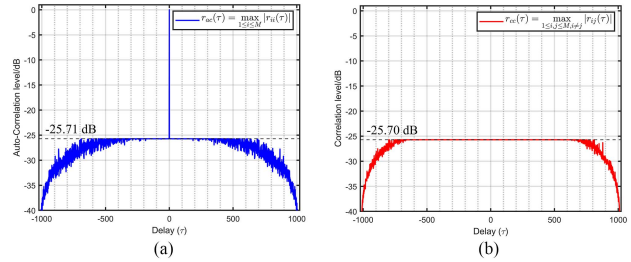

Fig. 8. Maximum value of each time lag for correlation functions obtained by the GD-based algorithm under $M = 128$. (a) Auto-correlation. (b) Cross-correlation.

1.5 dB improvement, and the third best ISL which is only 0.25 and 0.1 dB higher than the Welch lower bound and the best ISL, respectively. Overall, we can assert that the proposed method demonstrates favorable comparisons to the conventional algorithms, concerning PSL and ISL suppression.

The GD-based algorithm can also be applied to design large-scale waveform sets. The ISL achieved by the GD-based algorithm averages a 0.3 dB distance from the lower bound under $M = 4$, whereas only 0.01 dB under $M = 128$. The Primal–Dual optimizes PSL to -21.98 dB under $M = 128$, $N = 1024$, WeBEST-v optimizes to -24.3 dB, while the GD-based algorithm can optimize PSL to -25.7 dB, resulting in an advantage of 1.4 dB.

The maximum value of each time lag for auto-correlation and cross-correlation functions of the waveform set designed by the GD-based algorithm is shown in Fig. 8 under $M = 128$ and $N = 1024$. Using PSL as the optimization objective minimizes sidelobes of the correlation function. Although the amplitudes of the correlation functions within a small range of time lags are nearly identical between Fig. 8, the sidelobes of the correlation function remain lower for larger time lags. Balancing sidelobe levels at different time lags, PSL still holds the potential for further reduction.

E. Weighted PSL Optimization

The previous subsections discuss the suppression of all correlation sidelobes. According to (34), only different weights for each correlation sidelobes need to be set when using GD-based algorithm to suppress sidelobes in specific regions of interest. This subsection examines weighted correlation suppression performance of the GD-based algorithm under different waveform set sizes and compares it with multi-WeCAN and MM-WeCorr.

Under $M = 3$ and $N = 256$, assuming that only the correlation functions within time lags between 51 and 80 need to be concerned. To tackle this problem, we can set the weights of GD-based algorithm as

$$w_{ij}(\tau) = \begin{cases} 1, & 51 \le \tau \le 80,\ 1 \le i \le M,\ 1 \le j \le M, \\ 0, & \text{others}. \end{cases} \tag{43}$$

Only correlation functions with a corresponding weight of none zero will be suppressed. The auto-correlation and

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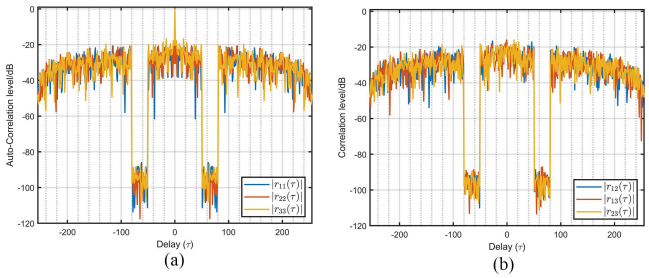

Fig. 9. Correlation functions obtained by the GD-based algorithm under $M = 3$ and $N = 256$ and the interested correlation functions within time lags between 51 and 80. (a) Auto-correlation. (b) Cross-correlation.

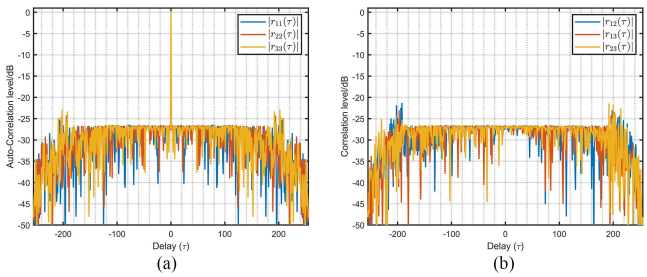

Fig. 10. Correlation functions obtained by the GD-based algorithm under $M = 3$ and $N = 256$ and the interested correlation functions within time lags between 0 and 191. (a) Auto-correlation. (b) Cross-correlation.

cross-correlation of the waveform set designed by the GD-based algorithm are shown in Fig. 9, where $M = 3$ and $N = 256$. The GD-based algorithm suppresses the correlation function within time lags between 51 and 80 to approximately -95 dB, while multi-WeCAN can optimize to about -50 dB and MM-WeCorr can optimize to -175 dB. Although the GD-based algorithm is not as effective as MM-WeCorr, it still has the value of practical application. Fig. 10 shows the auto-correlation and cross-correlation of the waveform set designed by the GD-based algorithm under $M = 3$ and $N = 256$, and the interested correlation functions within time lags between 0 and 191. As the time lags of concern gradually increase, the peak value of the correlation sidelobes within the time lags gradually increases, and is close to the PSL of the waveform set designed by suppressing all correlation sidelobes. Therefore, the time lags of concern and the target peak correlation sidelobes can be weighed to meet different practical needs.

F. Computational Time

In this subsection, the computational times of the algorithms proposed in this article and their comparison with other algorithms aimed at minimizing PSL are discussed.

Keeping the waveform set size and number of iterations $D = 20\,000$ unchanged, which is also the stopping criteria, we performed 10 experiments and recorded the average computational time.

Table I illustrates the computational time of algorithms, including Algorithm 1, an implementation using analytic GD on CPU, and Algorithm 2, a neural network-based implementation accelerated by GPU. Comparing the two different implementation methods of the GD-based algorithm, it can be observed that the acceleration effect of the neural network framework is significant. After inserting the performance tracking module inside the neural network, we found that the time-consuming process of calculating the gradients through automatic differentiation is comparable to that of calculating the loss function, while the direct calculation of gradients is approximately eight times more time-consuming than calculating the loss function, which is analyzed in Appendix-B with further details.

TABLE I
Computational Time

(M, N)	Algorithm 1 on CPU	Algorithm 2 on GPU
(4,1024)	41.14 s	28.70 s
(10, 256)	64.08 s	28.74 s
(10,1024)	314.41 s	29.57 s
(128,1024)	65868.05 s	398.42 s
(256,1024)	328514.60 s	1571.25 s

The neural network-based implementation not only computes gradients faster than the implementation using analytic gradient but also leverages GPU acceleration for parallel operations effortlessly. When the waveform set is small, data exchange between CPU and GPU accounts for most of the computational time, as shown in the first three rows of Table I. The acceleration effect of GPU is particularly pronounced when designing large-scale waveform sets.

The comparison of GD-based methods with other methods is carried out in the following and summarized in Table II. In this case study, we curated a list of waveform design algorithms aimed at minimizing PSL, including the GD-based algorithms (Algorithm 1 on CPU and Algorithm 2 on GPU), WeBEST-v, Primal–Dual, p-MM, and CON. Different from Table I, Table II used PSL as the stopping criteria, while Table I used the number of iterations as the stopping criteria. Due to the different extreme (best) PSLs achievable by different algorithms, the stopping criteria were set to PSLs that are about 1 dB higher than the lowest PSL obtained by the GD-based algorithm, as shown in the second column of Table II. However, some algorithms still could not optimize to the objective PSL in some cases. For these cases, the effective computational time could not be determined and "n/a" was used in Table II. Various waveform set sizes were studied. In each case, keeping the waveform set size and the objective PSL unchanged, we performed 10 experiments and recorded the average computational time. The initial sequence set was generated randomly as in the previous subsection for all algorithms.

Based on the results in Table II, it can be observed that for each (M, N) pair, the GD-based algorithm generally exhibits the best performance among all the algorithms. When N is less than 1024, Algorithm 1, which runs on a CPU, outperforms the GPU-based Algorithm 2. This is primarily due to the time overhead involved in data transfer

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TABLE II
Comparisons of Computational Times With Other PSL Algorithms ("n/a" Means the Algorithm Cannot Meet the Objective PSL)

(M, N)	Objective PSL	Algorithm 2 on GPU	Algorithm 1 on CPU	WeBEST -v	CON	Prime–dual	p-MM
(4, 64)	−18 dB	0.74 s	0.29 s	0.81 s	0.85 s	1.05 s	2.71 s
(4, 256)	−24 dB	2.04 s	1.37 s	4.63 s	9.05 s	11.91 s	116.86 s
(4, 1024)	−30 dB	2.61 s	7.98 s	10.65 s	n/a	n/a	n/a
(10, 1024)	−28 dB	15.27 s	65.14 s	72.37 s	n/a	n/a	n/a
(128, 1024)	−25 dB	257.86 s	23338.14 s	n/a	n/a	n/a	n/a

between the CPU and GPU, as well as neural network initialization. However, once N reaches 1024, the GPU's acceleration advantage becomes evident, with Algorithm 2 surpassing Algorithm 1 in computation efficiency. This difference is particularly pronounced when M is 128, where the CPU-based Algorithm 1 becomes impractically slow, requiring several hours to complete, while the GPU-based Algorithm 2 finishes the optimization in just a few hundred seconds.

As the set size increases, CON, Prime–Dual, and p-MM fail to converge to the specified objective PSL values, as indicated by "/" in Table II. Both the GD-based algorithm and WeBEST-v perform well when optimizing large waveform sets, except the case with $M = 128$ where WeBEST-v fails. The computational time of the GD-based algorithm before GPU acceleration, i.e., Algorithm 1 on CPU is comparable to that of WeBEST-v while Algorithm 2 on GPU is faster than WeBEST-v, especially for large waveform sets. For example, under $M = 4$ and $N = 1024$, they all successfully optimize the PSL to -30 dB. Algorithm 1 took 7.98 s and WeBEST-v took 10.65 s, both running on a CPU, while Algorithm 2 took 2.61 s. Additionally, under $M = 128$ and $N = 1024$, only the GD-based algorithm can optimize the PSL to -25 dB, completing the task in 257.86 s with GPU acceleration. The objective function in WeBEST-v is majorized by local approximate functions. The difference in approximation may be the reason why its PSL is slightly higher than that of the GD-based algorithm in large set sizes.

All algorithms listed in the table can meet the target PSL within a few seconds for smaller set sizes. In particular, under $M = 4$ and $N = 64$, the computational times are about 1 s for all algorithms. Thus, real-time optimization of small-scale waveform sets is feasible on resource-constrained airborne platforms, while large-scale waveform sets optimizing on resource-rich ground-based radar systems still take several hundred seconds.

Based on the above analysis, the GD-based algorithm demonstrates superior performance in terms of PSL optimization, computational efficiency, and scalability to large set sizes, as compared to the current methods.

VI. CONCLUSION

Motivated by neural networks, this article presents a GD-based algorithm for the design of orthogonal phase-coded waveform sets. The proposed method utilizes a first-order GD algorithm to minimize an objective function approximated by the LSE function. Moreover, this algorithm clarifies the connection with neural networks, making it compatible with mature software frameworks and significantly reducing programming effort. Numerical results highlight the algorithm's ability to optimize large-scale waveform sets, achieving lower PSL values by 1–2 dB compared to state-of-the-art algorithms while maintaining ISL close to its lower bound. The proposed algorithm demonstrates remarkable stability while also offering practical applicability due to its fast optimization speed.

APPENDIX A

According to [27], it is convenient to express the derivatives of real-valued function $f(x)$ with respect to the complex variable x as

$$\nabla_x f(x) = \lim_{\delta \to 0} \frac{f(x+\delta) - f(x)}{\delta} + j\frac{f(x+j\delta) - f(x)}{\delta}. \tag{44}$$

The partial derivative of $|r_{ij}(\tau)|^2$ with respect to $x_u(v)$ needs to be discussed in three cases with respect to the values of i, j, and u.

The first case is $i = u$ and $j \neq u$

$$\frac{\partial |r_{ui}(\tau)|^2}{\partial x_u(v)} = r_{ui}^*(\tau) x_u(v+\tau). \tag{45}$$

The second case is $i \neq u$ and $j = u$

$$\frac{\partial |r_{iu}(\tau)|^2}{\partial x_u(v)} = r_{iu}(\tau) x_u(v-\tau). \tag{46}$$

The third case is $i \neq u$ and $j = u$

$$\frac{\partial |r_{uu}(\tau)|^2}{\partial x_u(v)} = r_{uu}(\tau) x_u(v+\tau) + r_{uu}(\tau) x_u(v-\tau). \tag{47}$$

Therefore, the partial derivative of LSE_t with respect to $x_u(v)$ can be simplified to

$$\begin{aligned}
&\frac{\partial \mathrm{LSE}_t}{\partial x_u(v)} \\
&= \sum_{i=1}^{M} \sum_{j=1}^{M} \sum_{\tau=-N+1}^{N-1} w_{ij}(\tau) \frac{\partial \mathrm{LSE}_t}{\partial \left|r_{ij}(\tau)\right|^2} \frac{\partial \left|r_{ij}(\tau)\right|^2}{\partial x_u(v)} \\
&= \sum_{\substack{i=1 \\ i \neq u}}^{M} \sum_{\tau=-N+1}^{N-1} w_{ui}(\tau) \frac{\partial \mathrm{LSE}_t}{\partial |r_{ui}(\tau)|^2} r_{ui}^*(\tau) x_u(\tau+v)
\end{aligned}$$

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$$+\sum_{\substack{i=1\\i\neq u}}^{M}\sum_{\tau=-N+1}^{N-1} w_{iu}(\tau)\frac{\partial \mathrm{LSE}_t}{\partial |r_{iu}(\tau)|^2} r_{iu}(\tau)x_u(v-\tau)$$
$$+\sum_{\tau=-N+1}^{N-1} w_{uu}(\tau)\frac{\partial \mathrm{LSE}_t}{\partial |r_{uu}(\tau)|^2}(r_{uu}(\tau)x_u(v+\tau)$$
$$+ r_{uu}(\tau)x_u(v-\tau)). \tag{48}$$

Although the partial derivative is discussed in three cases, the expression of the third case shows that it is actually a special case of the first two cases. The formula can be further simplified to

$$\frac{\partial \mathrm{LSE}_t}{\partial x_u(v)} = \sum_{i=1}^{M}\sum_{\tau=-N+1}^{N-1} w_{ui}(\tau)\frac{\partial \mathrm{LSE}_t}{\partial |r_{ui}(\tau)|^2} r_{ui}^*(\tau)x_u(\tau+v) + \sum_{i=1}^{M}\sum_{\tau=-N+1}^{N-1} w_{iu}(\tau)\frac{\partial \mathrm{LSE}_t}{\partial |r_{iu}(\tau)|^2} r_{iu}(\tau)x_u(v-\tau). \tag{49}$$

Taking advantage of the conjugate symmetry of the correlation function, i.e.,

$$\left|r_{ij}(\tau)\right|^2 = \left|r_{ji}^*(-\tau)\right|^2. \tag{50}$$

Consequently, the final expression of the partial derivative of LSE_t with respect to $x_u(v)$ is

$$\begin{aligned}\frac{\partial \mathrm{LSE}_t}{\partial x_u(v)} &= 2\sum_{i=1}^{M}\sum_{\tau=-N+1}^{N-1} w_{ui}(\tau)\frac{\partial \mathrm{LSE}_t}{\partial |r_{ui}(\tau)|^2} r_{ui}^*(\tau)x_u(\tau+v)\\ &= 2\sum_{i=1}^{M}\sum_{\tau=-N+1}^{N-1} (w_{ui}(\tau)\frac{\partial \mathrm{LSE}_t}{\partial |r_{ui}(\tau)|^2} r_{ui}(\tau))^* x_u(\tau+v)\\ &= 2\sum_{i=1}^{M} h_{ui}(v)\end{aligned} \tag{51}$$

where

$$h_{ui}(v) = \sum_{\tau=-N+1}^{N-1} g_{ui}^*(\tau)x_u(\tau+v) \tag{52}$$

and

$$g_{ui}(\tau) = w_{ui}(\tau)\frac{\partial \mathrm{LSE}_t}{\partial |r_{ui}(\tau)|^2} r_{ui}(\tau). \tag{53}$$

Then, the vector form of LSE_t with respect to $x_u(v)$ can be expressed as

$$\nabla_{\mathbf{x}_u}\mathrm{LSE}_t = 2\sum_{i=1}^{M}\mathbf{h}_{ui}(\tau:\tau+N). \tag{54}$$

APPENDIX B

The computational effort required for gradient from (15) can be expressed approximately as

$$\Psi_g(M,N) = 3M^2(2N-1)\log(2N-1) + 3M^2(3N-2)\log(3N-2) + 2M^2(2N-1) + MN \tag{55}$$

where the first two items correspond to the correlation accelerated by FFT. One calculates the correlation function of the waveform set, and the other computes the correlation function of $\mathbf{g}_{ui}$ and $\mathbf{x}_u$. The last two terms involve weighting and a Hadamard product introduced during the gradient computation process. The cost function requires the same computational effort as the first term of $\Psi_g(M,N)$, which can be expressed approximately as

$$\Psi_c(M,N) = 3M^2(2N-1)\log(2N-1). \tag{56}$$

Calculate the ratio of $\Psi_g(M,N)$ to $\Psi_c(M,N)$, we can get

$$\Psi_g(M,N)/\Psi_c(M,N) \approx 8.5. \tag{57}$$

Therefore, the direct calculation of gradients is approximately eight times more time-consuming than calculating the loss function.

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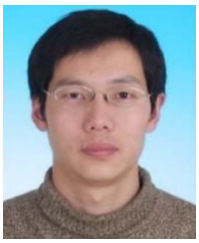

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